



Graphing Linear Inequalities on a Number Line

Algebra Worksheet · Grade 7-9

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Learning Objectives

- Graph simple linear inequalities on a number line using open and closed circles
- Solve one-variable linear inequalities using inverse operations
- Apply the rule of flipping the inequality symbol when multiplying or dividing by a negative number

Solve each inequality for x , then describe its graph on a number line including the type of circle (open or closed) and direction of shading.

1. Graph the inequality on a number line. State the type of circle and direction of shading.

$$x > 5$$

Answer: _____

2. Graph the inequality on a number line. State the type of circle and direction of shading.

$$x \leq 5$$

Answer: _____

3. Solve the inequality and describe its graph.

$$2x + 3 > 9$$

Answer: _____

4. Solve the inequality and describe its graph.

$$5 - 2x \geq -21$$

Answer: _____

5. Solve the inequality and describe its graph.

$$3x - 7 < 8$$

Answer: _____

6. Solve the inequality and describe its graph.

$$-4x + 6 \leq 22$$

Answer: _____

7. Solve the inequality and describe its graph.

$$\frac{x}{2} + 4 > 7$$

Answer: _____



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8. Solve the inequality and describe its graph.

$$-3x - 5 \geq 10$$

Answer: _____

9. Solve the inequality and describe its graph.

$$4(x - 2) < 12$$

Answer: _____

10. Solve the inequality and describe its graph.

$$7 - x \leq 2$$

Answer: _____



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Remind students that the inequality symbol flips when both sides are multiplied or divided by a negative number, and that 'greater than or equal to' / 'less than or equal to' use a closed (shaded) circle.

Solutions

1. Graph the inequality on a number line. State the type of circle and direction of shading.

$$x > 5$$

- The variable is already isolated.
- Use an open circle at 5 because the inequality is strictly greater than.
- Shade to the right because x is greater than 5.

Answer: $x > 5$

2. Graph the inequality on a number line. State the type of circle and direction of shading.

$$x \leq 5$$

- The variable is already isolated.
- Use a closed (shaded) circle at 5 because the inequality includes equal to.
- Shade to the left because x is less than 5.

Answer: $x \leq 5$

3. Solve the inequality and describe its graph.

$$2x + 3 > 9$$

- Subtract 3 from both sides to get $2x$ greater than 6.
- Divide both sides by 2 to get x greater than 3.
- Graph with an open circle at 3 and shade to the right.

Answer: $x > 3$

4. Solve the inequality and describe its graph.

$$5 - 2x \geq -21$$

- Subtract 5 from both sides to get negative $2x$ greater than or equal to negative 26.
- Divide both sides by negative 2 and flip the inequality symbol.
- The result is x less than or equal to 13.
- Graph with a closed circle at 13 and shade to the left.

Answer: $x \leq 13$

5. Solve the inequality and describe its graph.

$$3x - 7 < 8$$

- Add 7 to both sides to get $3x$ less than 15.
- Divide both sides by 3 to get x less than 5.
- Graph with an open circle at 5 and shade to the left.

Answer: $x < 5$



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6. Solve the inequality and describe its graph.

$$-4x + 6 \leq 22$$

- Subtract 6 from both sides to get negative 4x less than or equal to 16.
- Divide both sides by negative 4 and flip the inequality symbol.
- The result is x greater than or equal to negative 4.
- Graph with a closed circle at negative 4 and shade to the right.

Answer: $x \geq -4$

7. Solve the inequality and describe its graph.

$$\frac{x}{2} + 4 > 7$$

- Subtract 4 from both sides to get x over 2 greater than 3.
- Multiply both sides by 2 to get x greater than 6.
- Graph with an open circle at 6 and shade to the right.

Answer: $x > 6$

8. Solve the inequality and describe its graph.

$$-3x - 5 \geq 10$$

- Add 5 to both sides to get negative 3x greater than or equal to 15.
- Divide both sides by negative 3 and flip the inequality symbol.
- The result is x less than or equal to negative 5.
- Graph with a closed circle at negative 5 and shade to the left.

Answer: $x \leq -5$

9. Solve the inequality and describe its graph.

$$4(x - 2) < 12$$

- Distribute the 4 to get 4x minus 8 less than 12.
- Add 8 to both sides to get 4x less than 20.
- Divide both sides by 4 to get x less than 5.
- Graph with an open circle at 5 and shade to the left.

Answer: $x < 5$

10. Solve the inequality and describe its graph.

$$7 - x \leq 2$$

- Subtract 7 from both sides to get negative x less than or equal to negative 5.
- Multiply both sides by negative 1 and flip the inequality symbol.
- The result is x greater than or equal to 5.
- Graph with a closed circle at 5 and shade to the right.

Answer: $x \geq 5$



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